

Generative AI and Labor Demand - Evidence from EU Online Job Markets

Georgi Demirev
Sofia University

July 16, 2026

Abstract

This letter measures the early impact of generative AI on labor demand using online job postings across the European Union from 2021 to 2025. Using the release of ChatGPT as a cut-off date and five occupation-level measures of AI exposure as a continuous treatment, we document a 19-29% relative decrease in job postings for the most exposed occupations, with the gap widening each quarter after the release. The result is robust across exposure measures and to occupation and country ablations, including the exclusion of the entire ICT sector, and is partially replicated on Australian data, where a decade-long pre-period shows no confounding pre-trends.

JEL Classification: O33, J23, J24

Keywords: artificial intelligence, labor demand, online job postings, difference-in-differences

1 Introduction

The impact of artificial intelligence (AI) on labor demand has been an area of intense study over recent years, yet the empirical record of initial studies has been surprisingly mixed. Firm surveys point to widespread adoption coupled with modest effects on employment (Yotzov et al., 2026), and administrative data has so far failed to uncover evidence of sizeable AI-caused unemployment (Humlum and Vestergaard, 2025). At the same time, researchers have documented meaningful contractions in freelance labor demand for writing- and coding-intensive tasks (Demirci et al., 2025; Hui et al., 2023). The empirical facts of the matter seem far from settled.

Much of the difficulty in reconciling these findings stems from deficiencies in the granularity and timeliness of the available data, especially outside the United States, which has been the focus of most of the literature. This letter bridges part of this gap by leveraging high-frequency data on online job postings across the EU compiled by CEDEFOP (Cedefop, 2021), combined with five distinct task-based measures of occupational exposure to AI. Using the release of ChatGPT in November 2022 as a cut-off date, we estimate a continuous difference-in-differences specification with country fixed effects, with high-exposure occupations serving effectively as the treatment group

and low-exposure occupations as the control group. Relative to occupations with low exposure levels, those at the highest levels of AI exposure experience a 19-29% decrease in the number of online job adverts following ChatGPT’s release, with the associated event studies pointing towards the impact increasing over time.

Because the post-ChatGPT period coincides with major macroeconomic shocks across the examined region, we validate the finding on a second, independent data source - the Internet Vacancy Index (IVI) covering Australian online job markets. The two most recent exposure measures, including the only one derived from observed AI usage, recover comparable estimates there, and the decade-long pre-period available in the Australian data shows no pre-trends that would suggest a reversion to earlier labor market dynamics.

2 Data

The data on online job vacancies is taken from the Skills Online Vacancy Analysis Tool for Europe (Skills OVATE), developed jointly by CEDEFOP and Eurostat. The dataset contains information about more than 100 million job vacancies published on online job boards across the European Union and the UK, classified by occupation according to the ISCO/ESCO system. The data is updated quarterly, with each release covering the preceding twelve months; our sample spans Q4 2021 to Q2 2025. The analysis is carried out at the country by ESCO level-3 occupation level.¹

Occupational AI exposure is measured using five separate publicly available indices: Felten et al. (2018), Webb (2022), Eloundou et al. (2023), Demirev (2026), and Handa et al. (2025). The five measures capture different aspects of the overlap between AI capabilities and job skills - projected improvements in AI benchmarks, patented innovations, marketed AI products, a frontier model’s own assessment of task automatability, and revealed usage of a deployed AI system, respectively.² To ensure that the findings are not contingent on using a specific measure of exposure, all results are reported separately for each of the five indices, standardized to cover the 0 to 1 range.

For the out-of-sample validation we use the IVI dataset published by Jobs and Skills Australia, which contains online vacancies across the eight Australian states and territories by ANZSCO occupation, mapped to the exposure scores via the official ABS correspondence. An advantage of this dataset is the longer time period it covers - from 2016 onward - allowing us to examine a much longer window for pre-trends, reaching back through the pandemic.

3 Empirical strategy

Denoting the average number of online job adverts for occupation i in country c in the quarters before and after the release of ChatGPT as $\bar{y}_{c,i}^{PRE}$ and $\bar{y}_{c,i}^{POST}$, the main specification is a long-

¹Online job openings are only a proxy of real labor demand: some firms may keep open positions even if not actively recruiting, while occupations that rely on more traditional recruiting channels may be underrepresented in the data.

²The usage-based score of Handa et al. (2025) is constructed from conversations occurring after the release of ChatGPT, so it is best read as a measure of revealed adoption rather than pre-determined exposure.

difference regression of the log change on AI exposure:

$$\Delta \log \bar{y}_{c,i} = \delta_0 + \xi_c + \delta X_i + \varepsilon_{c,i} \quad (1)$$

where ξ_c are country fixed effects and X_i is one of the five exposure measures, treated as a continuous treatment. Standard errors are clustered at the country level. Cells with 20 or fewer adverts in either period are excluded, leaving roughly 2,788 country-occupation pairs. To study the time profile of the effect, we also estimate an event-study specification with country-occupation and time fixed effects:

$$\log y_{i,c}^t = \gamma_t + \alpha_{c,i} + \delta^t X_i \times \mathbf{I}(t) + \varepsilon_{i,c}^t \quad (2)$$

The same two specifications are estimated on the Australian data, with states and territories playing the role of countries.

4 Results

Table 1 reports the estimates of Equation 1. In the European data (Panel A), all five exposure measures yield negative coefficients, significant at the 1% level, ranging from -0.205 to -0.339 . This means that an increase in AI exposure from the lowest to the highest possible level results in between 19% and 29% (one minus the exponents of the regression coefficients) decrease in the number of online job adverts relative to the baseline period. The estimates are remarkably stable: across leave-one-out ablations dropping one occupation or one country at a time, not a single estimate flips sign or loses significance at the 5% level. Excluding the entire ICT block - ICT service managers, professionals, and technicians - at once attenuates the implied decline to 15-21%, but again leaves all five coefficients negative and significant.

Table 1: Impact of AI Exposure on Online Job Ads

	Demirev	Felten	Webb	Eloundou	Anthropic
<i>Panel A: European Union (Skills OVATE)</i>					
AI Exposure	-0.211*** (0.055)	-0.212** (0.058)	-0.205*** (0.043)	-0.289*** (0.066)	-0.339*** (0.054)
Observations	2,788	2,785	2,776	2,785	2,785
<i>Panel B: Australia (IVI)</i>					
AI Exposure	-0.152*** (0.025)	0.117* (0.045)	0.072 (0.046)	-0.018 (0.046)	-0.357*** (0.036)
Observations	1,197	1,197	1,197	1,197	1,197

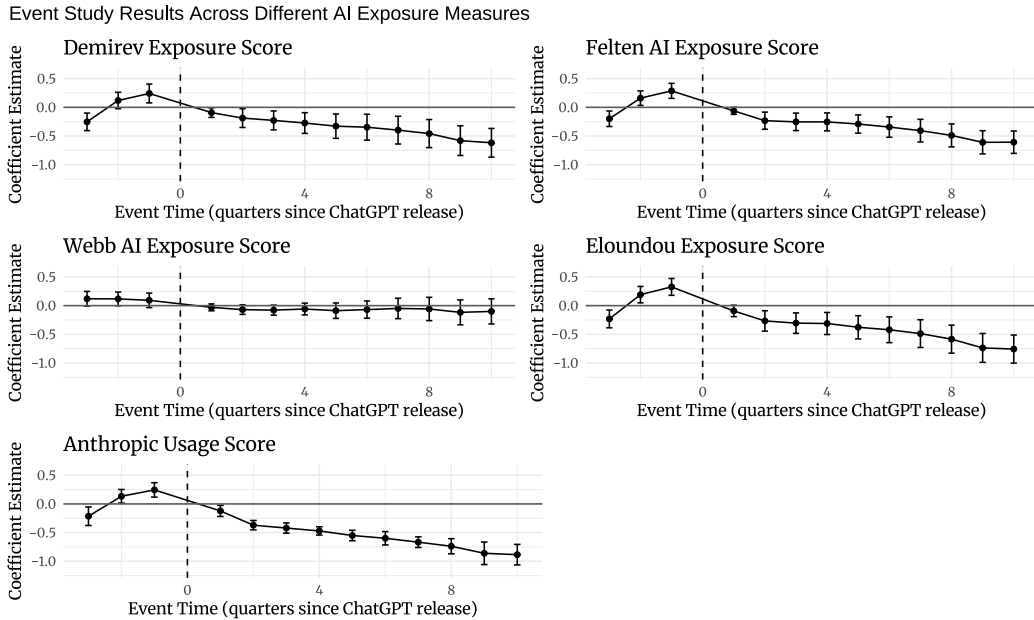
Dependent variable is $\Delta \log(\text{OJA})$. Country (state) fixed effects included.

Standard errors clustered at the country (state) level in parentheses.

* $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

Figure 1 presents the corresponding event studies. For four of the five measures, the coefficients turn significantly negative from the second quarter after the event date and grow in absolute magnitude with each successive period, reaching between 46% and 59% relative decrease by mid-2025; Webb (2022), a pre-LLM measure oriented towards earlier AI technologies, is the exception, with all periods insignificant. The pre-event coefficients, while much closer to zero, are significant for at least one period for the remaining four indices. The pre-event trend is, if anything, opposite in direction to the post-event one (rising rather than declining) - but a rising pre-event profile is also consistent with exposed occupations entering the treatment window at a cyclical peak. This concern is not innocuous: the 2022-2023 monetary tightening hit technology-intensive hiring particularly hard, and any factor affecting both occupational AI exposure and sensitivity to interest rate changes could bias the estimates. The stability of the estimates to dropping the ICT block noted above suggests the result is not simply a relabeling of the post-2022 technology hiring contraction, but subtler versions of the concern remain.

Figure 1: Event Study of ChatGPT’s Introduction against Number of OJA



The Australian data speaks directly to this concern. In Panel B of Table 1, the two most recent exposure measures - Handa et al. (2025) and Demirev (2026) - remain negative and significant, corresponding to a 14% to 30% relative reduction, bracketing the European estimates. The remaining three measures do not line up as cleanly: Eloundou et al. (2023) and Webb (2022) are insignificant, while Felten et al. (2018) is significant but positive. We therefore read the Australian evidence as a partial replication - supporting the direction of the headline result for the most up-to-date measures while leaving its universality across exposure measures an open question.³ Crucially, the Australian event studies, whose pre-period reaches back to 2016, show no systematic pre-trend for

³With only eight clusters, the Australian standard errors are not necessarily reliable in this specification.

the two replicating measures: the Handa et al. (2025) score is statistically indistinguishable from zero in every quarter after the onset of the pandemic, while the occupations deemed highly exposed by Demirev (2026) actually recorded a relative contraction in labor demand during the pandemic. This makes a mechanical reversion-to-the-mean from a pandemic-era swing an unlikely explanation for the post-ChatGPT decline.

Two of the exposure measures further decompose exposure into an "automation" and an "augmentation" component. In the Australian data both decompositions, and in the European data the usage-based one, attach the decline to the automation component (e.g., -0.316 , s.e. 0.044 , for Handa et al. (2025) in Australia) while the augmentation component enters positively - precisely the pattern the task-based framework predicts if the decline is driven by substitution rather than complementarity. This reinforces an interpretation of firms reducing their labor demand for automatable jobs while simultaneously increasing their demand for occupations in which workers are made productive by AI, although the two components are correlated within each measure and these coefficients should be read with caution.

In the companion working paper we further document that job skills most similar to AI capabilities are mentioned up to 12% less frequently within job descriptions, that industry-level labor and capital productivity show no measurable response - a combination consistent with what Acemoglu and Restrepo (2019) call "so-so automation" - and tentative descriptive evidence that the decline is concentrated in postings requiring either no experience or over ten years of experience.

5 Conclusion

Using high-frequency online job postings across the EU, we document a significant and growing relative decline in labor demand for AI-exposed occupations following the release of ChatGPT, robust across five exposure measures and partially replicated on Australian data. The magnitude of the estimates can already be seen as concerning for the prospects of labor in exposed occupations; however, it is worth remembering that these are only the initial effects as businesses are starting to adapt to the new technology. As AI continues to advance, monitoring labor market outcomes should remain an important task for policymakers and researchers concerned with job automation.

Declarations of interest: none.

References

- Daron Acemoglu and Pascual Restrepo. Automation and new tasks: How technology displaces and reinstates labor. *Journal of Economic Perspectives*, 33(2):3–30, May 2019. ISSN 0895-3309. doi: 10.1257/jep.33.2.3. URL <https://pubs.aeaweb.org/doi/10.1257/jep.33.2.3>.
- Cedefop. Skills Online Vacancy Analysis Tool for Europe (Skills-OVATE), 2021. URL <https://www.cedefop.europa.eu/en/tools/skills-online-vacancies>.
- Ozge Demirci, Jonas Hannane, and Xinrong Zhu. Who is AI replacing? the impact of generative AI on online freelancing platforms. *Management Science*, 2025. Also CESifo Working Paper 11276.
- Georgi Demirev. Ai product innovation and occupational exposure: automation and augmentation in commercial ai deployments. *Industry and Innovation*, pages 1–30, 2026. doi: 10.1080/13662716.2026.2623903. URL <https://doi.org/10.1080/13662716.2026.2623903>.
- Tyna Eloundou, Sam Manning, Pamela Mishkin, and Daniel Rock. Gpts are gpts: An early look at the labor market impact potential of large language models, August 2023. URL <http://arxiv.org/abs/2303.10130>. arXiv:2303.10130 [cs, econ, q-fin].
- Edward W. Felten, Manav Raj, and Robert Seamans. A method to link advances in artificial intelligence to occupational abilities. *AEA Papers and Proceedings*, 108:54–57, May 2018. ISSN 2574-0768, 2574-0776. doi: 10.1257/pandp.20181021. URL <https://pubs.aeaweb.org/doi/10.1257/pandp.20181021>.
- Kunal Handa, Alex Tamkin, Miles McCain, Saffron Huang, Esin Durmus, Sarah Heck, Jared Mueller, Jerry Hong, Stuart Ritchie, Tim Belonax, Kevin K. Troy, Dario Amodei, Jared Kaplan, Jack Clark, and Deep Ganguli. Which economic tasks are performed with AI? evidence from millions of Claude conversations, February 2025. URL <https://arxiv.org/abs/2503.04761>. Anthropic Economic Index, release 2025-03-27.
- Xiang Hui, Oren Reshef, and Luofeng Zhou. The short-term effects of generative artificial intelligence on employment: Evidence from an online labor market, July 2023. URL <https://ssrn.com/abstract=4527336>. Available at SSRN.
- Anders Humlum and Emilie Vestergaard. Still waters, rapid currents: Early labor market transformation under generative ai. Working Paper 33777, National Bureau of Economic Research, May 2025. URL <http://www.nber.org/papers/w33777>.
- Michael Webb. The impact of artificial intelligence on the labor market, 2022. URL https://www.michaelwebb.co/webb_ai.pdf.
- Ivan Yotzov, Jose Maria Barrero, Nicholas Bloom, Philip Bunn, Steven J Davis, Kevin M Foster, Aaron Jalca, Brent H Meyer, Paul Mizen, Michael A Navarrete, Pawel Smietanka, Gregory

Thwaites, and Ben Zhe Wang. Firm data on ai. Working Paper 34836, National Bureau of Economic Research, February 2026. URL <http://www.nber.org/papers/w34836>.